Summary in Graph

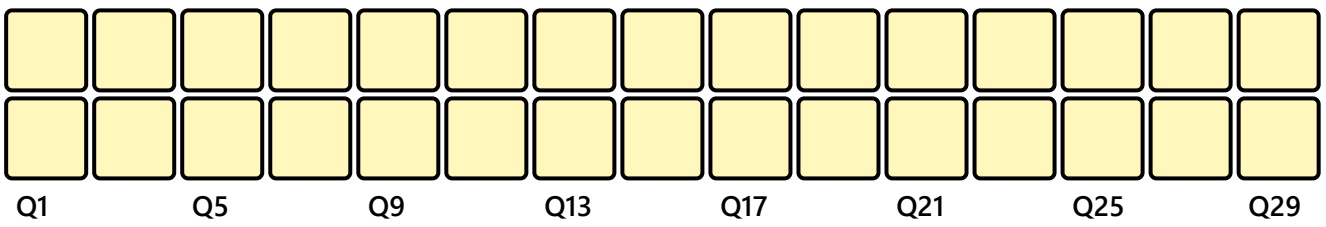
Exam Summary (GATE Overflow CS Test Series 2026 | Digital Logic | Shallow Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (3) out of 26 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Which of the following boolean expressions does not match the equivalence of the others?

A. $(x' \odot y) \cdot (x' \odot y')$

B. $(x' \cdot y) \cdot (x' \cdot y')$

C. $(x' \oplus y) \cdot (x' \oplus y')$

D. $(x' \cdot y) + (x' \cdot y')$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Which of the following Set of Boolean functions is functionally complete?

A. {XNOR, 0, AND}

B. {XOR, 1, AND}

C. {XNOR, 0, OR}

D. All of the above.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

In which of the following signed number representations do we have an equal number of negative numbers and positive numbers?

A. 1s Complement Representation
B. 2s Complement Representation
C. Sign Magnitude Representation
D. Both A and C

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

The decimal number 45.625 can be expressed in scientific notation as:

A. 1.01101101×2^5
B. 1.0111001×2^5
C. 1.011101×2^6
D. 1.011101×2^4

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Which of the following statement(s) is/are false?

A. AND and NAND are associative.

B. NOR and XNOR are associative.

C. 1-input NOR is not the complement of itself.

D. All the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Which of the following options represents the simplified form of the Boolean expression $(A + B'C')' + (AB)'$?

A. $A' + B'$

B. B

C. $A'B'$

D. $A'C$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Digital Logic

Consider the following assertions regarding the operations of Boolean addition. Which of these statements are valid? (Select all that apply)
1. Adding two false values results in false, and adding a false and a true value results in true.
2. The sum of two true values is always true, regardless of any other conditions.
3. The addition of true and false values yields a false result.
4. The addition of two false values yields a true result, while true and false yields a false result.

A. Statement 1 is correct.

B. Statement 2 is correct.

C. Statement 3 is correct.

D. Statement 4 is correct.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Digital Logic

If $A \oplus B = C$ (where $\oplus$ represents the XOR operation). Which of the following statements are false? (Mark all the appropriate choices)

A. If you XOR C with A, the result is B.
B. If you XOR C with B, the result is A.
C. The expression $A \oplus B \oplus C$ evaluates to 1.
D. The result of $A \oplus A$ equals C.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Digital Logic

Which of the following multiplier inputs is most likely to generate the greatest number of subtraction operations (considering subtraction as $2's$ complement addition) in $Booth's$ multiplication method?

A. 111000
B. 000011
C. 101111
D. 010101

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Numerical Type | Award: 1 | Penalty: 0 | Digital Logic

Assume that the propagation delay in each gate in a 16-bit carry look-ahead adder (using AND, OR, and XOR gates with up to 4 inputs) is $0.5ns$. The time taken in nanoseconds to perform a 16-bit addition is ____.

Your Answer: | Correct Answer: 3:3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Which of the following decimal number causes a loss of precision when represented in IEEE-754 single precision representation?

A. 524288×524288

B. 16777217

C. 100100100

D. Both B & C

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Which of the following 32-bit binary streams correctly represents the NaN (Not a Number) value in IEEE-754 single-precision format?

A. $\underbrace{0}_{\text{sign}}\ \underbrace{11111111}_{\text{exponent}}\underbrace{00000000000000000000000}_{\text{mantissa}}$

B. $\underbrace{0}_{\text{sign}}\ \underbrace{11111111}_{\text{exponent}}\underbrace{00000000000000000000001}_{\text{mantissa}}$

C. $\underbrace{1}_{\text{sign}}\ \underbrace{11111111}_{\text{exponent}}\underbrace{00000000000000000000000}_{\text{mantissa}}$

D. $\underbrace{0}_{\text{sign}}\ \underbrace{11111110}_{\text{exponent}}\underbrace{00000000000000000000001}_{\text{mantissa}}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Let C represent the smallest positive normalized value in the IEEE-754 single-precision floating-point format, and let D denote the smallest positive normalized value in the IEEE-754 double-precision format (both disregarding subnormal values and other special cases). What is the approximate value of the ratio D/C?

A. 10^{-38}
B. 10^{-154}
C. 10^{-308}
D. 10^{-270}

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Match the following Special values to their correct IEEE-754 double precision format:

Special Values	IEEE-754 Representations
1.Zero	$P.$ $\underbrace{0}_{\text{sign}}\ \underbrace{11111111111}_{\text{exponent}}\underbrace{0000000000000000000000000000000000000000000000000000}_{\text{mantissa}}$
2.Infinity	$Q.$ $\underbrace{0}_{\text{sign}}\ \underbrace{00000000000}_{\text{exponent}}\underbrace{0000000000000000000000000000000000000000000000000000}_{\text{mantissa}}$
3.Denormalized	$R.$ $\underbrace{0}_{\text{sign}}\ \underbrace{00000000000}_{\text{exponent}}\underbrace{0000000000000000000000000000000000000000000000000000}_{\text{mantissa}}$
4.Not a Number (NaN)	$S.$ $\underbrace{0}_{\text{sign}}\ \underbrace{11111111111}_{\text{exponent}}\underbrace{0000000000000000000000000000000000000000000000000000}_{\text{mantissa}}$

A. 1-P, 2-Q, 3-R, 4-S
B. 1-Q, 2-P, 3-S, 4-R
C. 1-Q, 2-P, 3-R, 4-S
D. 1-S, 2-R, 3-Q, 4-P

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

If the $2's$ complement notation of a positive number x is denoted as M, and the $2's$ complement notation of its negative counterpart $-x$ is represented as N, what will be the result of the operation $M \text{ NAND } N$ (where NAND signifies the NAND operation)?

A. All bits 0s
B. All bits 1s
C. 1 less than a power of 2
D. An even number, if x is odd & vice-versa.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Consider the following Boolean function:

$$F = AB' + AC + B'C'$$

It is found that the circuit input combination $A = 0, B = 1$ can never occur. The simplified expression for F using the proper $don't$-care conditions is:

A. $F = AB' + AC$
B. $F = A + B'$
C. $F = B'C'$
D. $F = AC + B'C'$

Which option correctly simplifies the function considering the $don't$-care condition?

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

A boolean function H together with the $don't$ care condition d are given below:
$H(P, Q, R, S) = \sum(0, 3, 5, 7, 9, 11)$
$d(P, Q, R, S) = \sum(1, 2, 4, 6, 10)$
The simplified form of H can be expressed in the sum of minterms as

A. $\sum(0, 1, 2, 3, 4, 5, 6, 7, 9, 11)$
B. $\sum(0, 1, 2, 3, 4, 5, 6, 7, 9, 10, 11)$
C. $\sum(1, 3, 5, 7, 9, 11)$
D. $\sum(0, 3, 5, 7, 9, 11)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

Let $F(X, Y, Z, W) = \sum(1, 3, 7, 10, 11)$. If the number of prime implicants and essential prime implicants of this function are represented by p and q respectively, what is the value of $3^p \times 5^q$?

Your Answer: | Correct Answer: 10125:10125 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

A number is expressed with 16 digits in decimal format. If we want to convert this number to binary, how many digits will it take if we represent the number using the smallest power of 2 that is greater than or equal to the decimal number?.

Your Answer: | Correct Answer: 54:54 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

If there are 4 distinct Boolean variables, and you want to choose k variables from these to form a Boolean function, what is the total number of different Boolean functions that can be formed using exactly 2 of these variables?

Your Answer: | Correct Answer: 96:96 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Which of the following statements is/are Correct regarding the dual of the boolean function.
P: The dual of a sum of products expression is always a product of sums expression.

Q: When converting a product of sums expression to its dual, the number of terms may increase or decrease based on the structure of the original expression.

R: The dual of a function does not change if you negate the entire function before finding its dual.

S: If two functions are duals of each other, their truth tables will have the same number of rows and may have same output values for some input combinations.

A. R and S
B. P, R and S
C. Only R
D. P and S

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

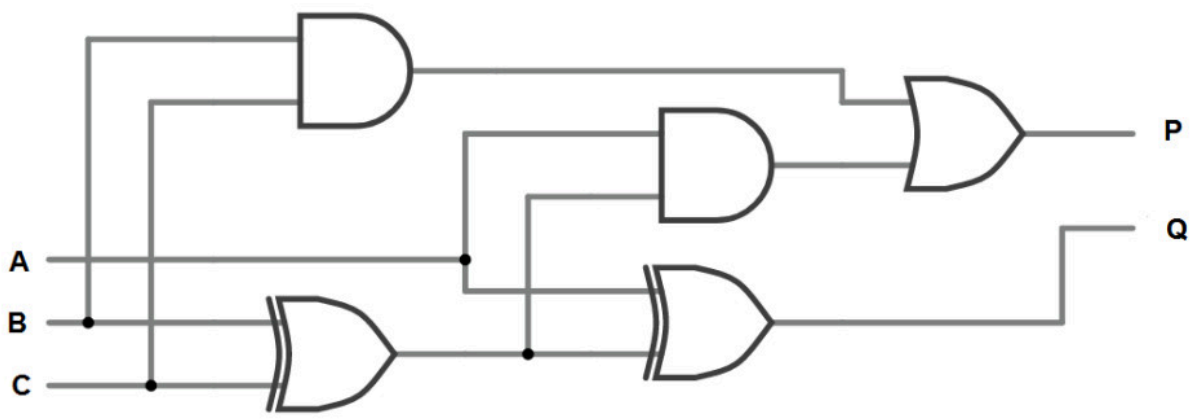
Q #22 Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

Two binary numbers are represented in fixed-point format as 1010.011 and 110.101. If the second number is subtracted from the first, what is the result in decimal form, rounded to the nearest integer?.

Your Answer: | Correct Answer: 4:4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Given the circuit diagram below, if $A = C = 1, B = 0$ then P and Q are respectively



A. $1, 0$
B. $0, 1$
C. $1, 1$
D. $0, 0$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 02sec | Discuss

Q #24 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The implicants being chosen for the circuit implementation are marked in the following four Karnaugh-maps. In which of them, a static-0 hazard is possible?.

A.

pq \ rs	00	01	11	10
00	0	1	1	0
01	1	1	0	0
11	0	1	1	1
10	0	1	1	0

B.

pq \ rs	00	01	11	10
00	0	1	1	0
01	1	1	0	0
11	0	1	1	1
10	0	1	1	0

C.

pq \ rs	00	01	11	10
00	0	1	1	1
01	0	0	1	1
11	0	0	0	1
10	0	1	1	1

D.

pq \ rs	00	01	11	10
00	0	1	1	1
01	0	0	1	1
11	0	0	0	1
10	1	1	1	1

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The 16-bit $2's$ complement representation and the hexadecimal representation after finding the $2's$ complement of the decimal number 2024, respectively, are:

A. $1111\ 1000\ 0001\ 1000,$ `0xF818`
B. $0000\ 0111\ 1111\ 1000,$ `0x07F8`
C. $0000\ 1000\ 1110\ 1000,$ `0x07E8`
D. $1111\ 1000\ 0111\ 1000,$ `0xF878`

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

The decimal and hexadecimal representation for the bit string 101001110011, represented in 16-bit $1's$ complement representation are:

A. $-1420,$ `0xFA73`
B. $1420,$ `0x57C`
C. $-1234,$ `0xF3DA`
D. $1234,$ `0x4D2`

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27

Numerical Type | Award: 2 | Penalty: 0 | Digital Logic

Let U denote the highest value achievable for an unsigned integer with a 5-bit binary representation, while V represents the highest signed integer that can be represented in 5 bits, utilizing $1's$ complement

representation, compute $U - V =$_____ .

Your Answer: | Correct Answer: 16:16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

In the context of floating-point representations, which octal encoding (according to the IEEE-754 single precision format) accurately represents the decimal value 5.75?

A. 0o10056000000
B. 0o40056000000
C. 0o40030000000
D. 0o42020000000

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Which of the following statements about denormalized numbers in IEEE-754 floating point representation is correct?

S1: Denormalized numbers allow representation of values closer to zero than normalized numbers can.
S2: The exponent field of denormalized numbers is set to zero, which represents the smallest possible exponent in the biased representation.
S3: Denormalized numbers can provide gradual underflow compared to normalized numbers.

A. Only $S1$
B. Both $S1$ and $S3$
C. Both $S2$ and $S3$
D. $S1$, $S2$ and $S3$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Digital Logic

Which of the following correctly describes the representation of a signed integer in $1's$ complement notation as a sum of powers of 2? A. $(b_{n-1}b_{n-2}\ldots b_2b_1b_0)_2 \equiv \sum_{i=0}^{n-1} b_i 2^i$ B. $(b_{n-1}b_{n-2}\ldots b_2b_1b_0)_2 \equiv -b_{n-1}2^{n-1} + \sum_{i=0}^{n-1} b_i 2^i$ C. $(b_{n-1}b_{n-2}\ldots b_2b_1b_0)_2 \equiv -b_{n-1}2^{n-1} + \sum_{i=0}^{n-2} b_i 2^i + b_{n-1}$ D. $(b_{n-1}b_{n-2}\ldots b_2b_1b_0)_2 \equiv -b_{n-1}2^{n-1} + 2^n + \sum_{i=0}^{n-2} b_i 2^i$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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